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Embodied energy and associated carbon emission of key building materials in Nepal

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Abstract
The number of concrete buildings in Nepal increased by 23.90 percent within the last decade. Life Cycle Assessment (LCA) of the buildings shows that Cement, Brick and Reinforcement steel are the three major building materials which account for about half of the total life cycle energy use and emission from the building materials. However, there is no national database for energy use and emissions from these building materials in Nepal. So, the study aims to evaluate energy use and its associated emissions in the production of these materials using the LCA framework and guidelines from ISO 14040: 2006 and ISO 14044: 2006. The data from embodied energy is based on the energy audits of 26 cement industries, 21 metal industries, and 27 brick industries sampled across the country. The study shows that the production of one tonne of cement accounts for 6051.07 MJ energy and is responsible for 739.49 $kgCO_2$-eq.; the production of 1000 pieces of standard size burnt brick from fixed chimney bull trench kiln accounts for 4124.56 MJ energy and 502.89 $kgCO_2$-eq. emission; and the production of one tonne of reinforcement steel accounts for 26,033.14 MJ energy and 2565.5 $kgCO_2$-eq emission. The major source of energy and emission in building material production is coal. A shift in energy sources from coal to hydroelectricity would reduce the energy-related emissions from the materials production. Also replacing high emission construction materials with locally available natural materials like stone, wood and bamboo could minimize the emissions from the built environment.

Keywords Brick · Carbon emission · Cement · LCA · Reinforcement steel

1 Introduction

Greenhouse gases (GHG) generated from anthropogenic sources mainly responsible for global warming and climate change [1]. In 2022, the International Energy Agency (IEA) reported an increase in global carbon dioxide equivalent (CO_2-eq.) emissions from escalating fuel combustion and industrial activities reaching 36.60 Gigatons [2]. Carbon dioxide (CO_2) emitted from fossil fuel combustion in industrial activities, commercial enterprises, power generation, transportation, and other sources has led to an exponential increase in atmospheric CO_2 levels to approximately 423 parts per million (ppm) [3], resulting in the trapping of long-wave radiation and a consequent rise in atmospheric temperatures

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[4]. Observations indicate 1.09 (0.95 to 1.20) °C global surface warming over the last decade (2011–2020) compared to the pre-industrial period (1850–1900) with larger increases over land with 1.59 (1.34 to 1.83)°C [5]. Under the 2015 Paris Agreement, countries committed to limiting global warming to "well below 2.0 °C" through mitigation measures. By the end of the century, global warming would still reach nearly between 2.4 to 2.80°C despite the Agreement's promises [6]. A worldwide coalition for carbon neutrality is presently in place to limit global warming to 1.5 °C. The countries involved in this coalition comprise over 65 percent of GHG emissions and over 70 percent of the global GDP, and they have committed to achieving net-zero carbon dioxide emissions [6, 7].

Buildings account for significant amounts of energy, water, and raw material consumption and emission of GHGs and waste [8]. Globally buildings account for 30 percent of the total raw material utilization, 25 percent of the water used, 12 percent of the land use, and 25 percent of the solid waste generated [5, 9]. Throughout a building's lifetime, both direct and indirect energy are used. Direct energy is used for installing and moving building materials, whereas indirect energy is used for extracting, processing, and producing construction materials [10]. The building operation and construction utilize 34 percent of the global final energy consumption and account for the emission of 40 percent of the energy-related CO_2 emission [7]. The CO_2 emission of the buildings is higher because of intensive energy utilization in raw material extraction, material production, and transportation which are mostly powered by fossil fuels [10]. Use of alternative building materials and techniques can cut CO_2 emissions by up to 90% during various phases of building [8].

Between 2011 and 2021, there is an increase in number of buildings in Nepal by 22.82 percent [11], while the percentage of structures with reinforced concrete roofs and cement-bonded brick/stone walls increased by 23.90 [11, 12]. This shows that along with increase in number of buildings in Nepal, there is also shift in building construction practice from traditional building to contemporary concrete buildings. This indicates that bricks, cement, and reinforcement steel are becoming the major building construction materials in Nepal. Also growing infrastructure development and urbanization are signs that developing nations are producing and utilizing more building materials [13].

The life cycle study of modern frame structure reinforced concrete buildings in Nepal shows that 49 percent of the life cycle energy use and 60 percent of CO_2-eq. emission is during material production and construction stage [14]. Cement, brick and reinforcement steel are the three most energy intense and emitting materials among the materials used in the building and these three materials utilizes 75 percent of the total energy use in material production and construction stage of the building and contributes 69 percent of emission [14]. Production of cement, brick, and reinforcement steel involves several production processes and equipment. The used equipment has a long service life, and fuel/energy utilization in different phases and forms which requires a holistic analysis for the determination of the embodied energy and CO_2 emission in the production. The industrial sector in Nepal utilizes different sources of energy out of which 48.15 percent of the total energy demand is fulfilled by coal, 16.83 percent by firewood, 12.75 percent by diesel, 9.98 percent by agriculture residue, and only 8.87 percent by electricity supplied from the national grid [15, 16]. Cement, brick, and steel industries are among the most significant sources of GHGs.

About 7 percent of all CO_2 emissions worldwide are attributed to the cement industry, which ranks second in CO_2 contributions and third in industrial energy use [17]. There is a projected 12 percent rise in global production of cement by 2050 [17], while the businesses are striving to reduce CO_2 emissions per unit through improvement of energy efficiency and shift to cleaner energy. In Nepal, coal and fossil fuels are basically used in manufacturing of cement while electricity is used in small fraction [15]. However, through its second Nationally Determined Contribution (NDC), Government of Nepal (GoN) aims to expand its hydroelectricity production to 15,000 MW by 2030 and reduce coal consumption in brick and cement industries [18].

Brick production, an age-old craft in Nepal, has played a crucial role in building temples and historical structures. The increasing urbanization has led to a substantial surge in brick demand. As per ICIMOD's factsheet of 2018 on the Brick sector in Nepal, there were 1349 brick kilns [19] of which 110 are located in the Kathmandu Valley [20]. These industries primarily rely on coal and manual labor. Brick kilns in Nepal typically yield between 15,000 and 50,000 bricks per day, predominantly through manual molding with a usual operational span of about six months, from December to June, as the monsoon starts from June and the brick moldings and drying is impacted by rain [21]. Brick kilns are significant contributors to air pollution in Nepal [22], with emissions varying based on kiln type, firing duration, technology, and fuel employed [23].

Another vital part of modern infrastructure and reinforced cement concrete building are steel and iron. They are the second most widely used building construction material contributing 7.2 percent of global GHGs emissions [24]. A quarter of the direct carbon dioxide emissions from all industrial sectors worldwide are attributed to the iron and steel industries, which are also large energy users and emitters [25]. Product design, cycle time, energy use, waste reduction, and equipment efficiency are some of the variables that influence the environmental emission of production iron and

steel production [26]. Iron billets are imported from India which are then rolled in hot rerolling mills to produce reinforcement steel.

There are country-specific energy use and emission factors for each construction material due to varying technological use, energy mix, and transportation of raw materials and fuels. The emission factors for specific fuels are used for the determination of the embodied CO_2-eq. of cement, bricks, and reinforcement steel. The energy use is determined based on the production process and source of energy used in the extraction, processing, and production of the building materials and their transportation. The total energy use and emission of the materials is determined to cover the complete stages of the materials production [23]. The emissions are divided into two categories direct and indirect. The direct emission from material production includes emission during the production process and fuel consumption. Indirect emission in the production of these materials is generated from the electricity consumed in the respective industries in the material production and the amount of fuel consumed in the handling and transportation of the materials [27]. The emission from the electricity mix of the country is determined by emissions from different sources of electricity and their emission factors.

Table 1 provides a summary of past studies and methodological approaches that have been employed to study the environmental impacts of different building materials. Table 1 provides a useful resource for comprehending the sustainability landscape in the construction sector by arranging information on materials such as steel, bricks, concrete, and wood products, together with their corresponding techniques and major results in different countries. Most of the studies conducted in different countries used the cradle-to-gate approach of Life Cycle Assessment (LCA) for determining the embodied energy and CO_2-eq. emission of the construction materials. Emission factors were employed for quantifying CO_2-eq. emission from construction materials. It was highlighted that direct emission contributes to higher CO_2-eq. emissions. Several authors discussed the possibilities of minimizing emissions by shifting energy sources from coal to electricity in steel, cement, and brick production. Brick drying and firing processes contribute the higher energy use and emission in brick production [23, 28]. The need for substitution of cement with alternatives or recycled materials to minimize emission was pinpointed [29–31].

LCA study requires the country specific data to accurately estimate the environmental impacts of a product or system. There are limited studies on the emission from the production of contemporary building materials in Nepal, despite their rapid increase in use. There are also no legitimate criteria and national standards in the country for industries to determine the overall emission of their product so there are no such holistic studies or data on the emission of the building construction materials. Similarly, there are no Environmental Product Declarations (EPDs) of the construction materials in Nepal. The LCA studies in Nepal and South Asian region use international database because of unavailability of local databases and the studies have emphasized the need of country specific database to minimize the uncertainties of the LCA of buildings and infrastructures in South Asian countries [13, 40].

Lack of national database and no provision of EPD on the building materials corroborates the need of study on embodied energy and emission from the contemporary building materials in Nepal. This shortcoming motivates the study of embodied energy and carbon emissions from the production of the contemporary building material in Nepal. This study aims to assess the embodied energy and CO_2-eq. emissions from the production of three construction materials viz. cement, brick, and reinforcement steel in Nepal. The study focused particularly on energy use and global warming potential (GWP) to understand the impact of the construction material on the most pressing issue of the twenty-first century: energy stress and climate change. This study will support in efforts to create a national database for the major contemporary building material and support future LCA studies on building and infrastructure. The outcome of the study will also help to promote the use of materials with low energy use and emission and promote the necessity of circularity in the building construction sector.

2 Methodology

Measurement of emission from all the sources in construction material production is impractical, so the most common approach to estimate the emission is to combine information on the extent of activities with the energy utilization and emission factors from different sources of energy and production processes. The activity data of the energy use throughout the life stage of production of the respective construction materials is multiplied by the emission factors to determine the emission [41]. LCA is a widely accepted and its framework is generally used to assess the embodied energy and embodied CO_2-eq. emission of the buildings and construction materials [10]. This study utilized the LCA framework

Table 1 Summary of environmental impact studies on different building construction materials

Authors	Material studied	Methodology/approach	Key findings
[32]	Steel	Assessment of emissions from the blast furnace, the basic oxygen furnace, and casting rolling, is done	The total emissions can be allotted into direct, upstream, and by-product emissions, where the biggest impacts of global warming potentials were from direct emissions which were 48% from the power plant and 22% from the blast furnace. The study shows that the GWP impact is 2.1 kg CO_2-eq per kg of hot rolled coil
[33]	Steel	LCA with cradle-to-gate approach	The study highlights the possibility of shifting energy sources from coal to electricity in steel production. The way of production of electricity plays a crucial role in the future and guides the emission from the steel industry
[34]	Steel	Life cycle inventory (LCI) study- with cradle-to-gate approach	The study represented over 99% of steelmaking technologies worldwide and covered over 26% of steel production by companies on a global basis. The study used primary steelmaking process steps plus boilers, compressors, water intake, effluents, stockpile emissions, and transportation of raw materials
[23]	Clay bricks	Analysis of life cycle carbon emissions and embodied energy	The use of sun-dried bricks instead of fired bricks can reduce CO_2 emission reduction of up to 5907 kg CO_2-eq. and 5305 MJ energy for every 1000 bricks
[28, 35]	Bricks and construction blocks	Analysis of energy-intensive processes, especially firing and drying	Firing and drying processes contribute to 87% of total energy employed; the firing process (800–2000°C) is the most impactful; choice of energy mix/fuel influences environmental impact; cogeneration reduces the power required by 10%
[28]	Adobe bricks	Comparison of environmental impact with traditional fired bricks	Adobe bricks are more environmentally friendly, with lower embodied energy and carbon; lack of firing process; performance study of operational use is challenging
[36]	Clay bricks with additives	Laboratory-scale manufacturing of clay bricks with bio-based additives	Additives can increase porosity, reduce thermal conductivity, and improve embodied components; environmental benefits observed; mechanical performance may be compromised
[37]	Traditional and alternative bricks	Review of environmental impacts with organic and inorganic additives	Use of recovery or reused waste materials for sustainability; the influence of recovered material on mechanical performances still under study
[38]	Ceramic bricks	Proposed methodology for "cradle to grave" EPD specific for ceramic bricks	Neglects the impact of use stage (B1–B4 modules); functional unit suggested is 1 ton or 1 m^3 for "cradle to gate" studies; operational stages not considered in EPDs
[39]	Cement	Calculation with emission factors	The study accessed the composition of samples from cement production in China and the raw material carbonate method to calculate process-related Emission Factors

adopted from ISO 14040: 2006 [42] and ISO 14044: 2006 [43] comprising four phases: Goal and Scope definition, Inventory analysis, Impact assessment, and Interpretation.

2.1 Goal and scope definitions

The aim of the study is to determine the embedded energy and its associated carbon emissions of the three major construction materials, i.e. cement, bricks and reinforcement in Nepal. The study used the LCA framework with cradle-to-gate approach to assess the energy utilization and carbon emissions associated with the energy use in the production of cement, brick, and reinforcement steel in Nepal using the ISO 14040:2006 framework [44]. However, the cradle-to-gate approach was used because most of the environmental product declarations (EPD) are based on EN15804: 2011 and it does not consider the end-of-life of the products [45]. The life cycle impacts of the building materials generally examine two impact categories: global warming potential (GWP) in kgCO_2-eq. and energy use in MJ which are investigated in the study [45].

Figure 1 presents the generalised system boundary considered in the study. At the factory's point of entry, the production of a particular amount or mass of material is referred to as the functional unit. The functional unit to produce cement and reinforcement steel is one tonne of cement and reinforcing bars, respectively. While for bricks, the functional unit is the production of one thousand fired clay bricks. Production processes involving loading and unloading of cement are done manually which is not accounted for in this study. Processes like watering the soil for brick production, brick moulding, and transportation involve humans and donkeys in Nepal which is not accounted for while determining energy utilization and CO_2 emission. The moulded clay bricks are sundried in Nepal before firing them in the kiln. The energy utilization and emission from the studied material is portrayed in Fig. 1.

2.2 Inventory analysis

The study for embodied energy is based on the secondary data from the energy audit conducted on 26 cement industries, 21 metal industries, and 27 brick industries sampled across the country conducted by Nepal Energy Efficiency Program (NEEP) [21]. The energy audit was conducted on energy intensity and energy consumption in the sampled industries across different parts of the country. The cement and reinforcement steel industries use the electricity from the national grid of Nepal Electricity Authority through 66/33/11 kV supply line, while the fossil fuels used in these industries are imported from India. Both cement and reinforcement steel industries have diesel generators for the backup power supply during power failure from the grid. The studied cement industries have annual production capacity of 3.6 million tonnes.

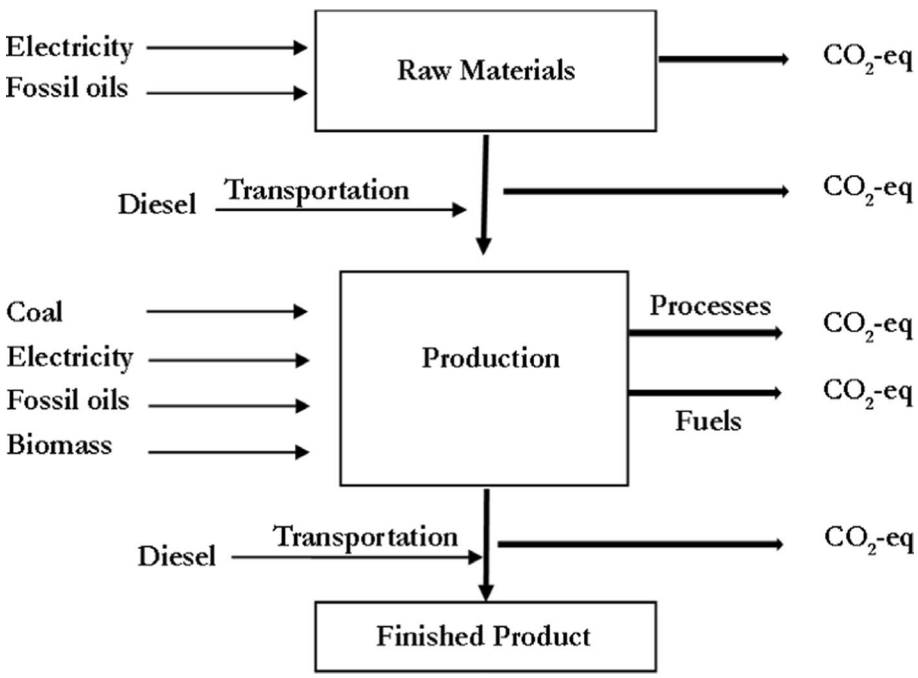


Fig. 1 Systems components considered for energy use and carbon emission analysis

The assessed industries produced 1.5 million tonnes of cement with consumption of 105.3 million kWh electricity and 3,836.6 million MJ energy during the period. The metal industries for reinforcement steel production utilize hot- rerolling and wire drawing process using imported cast billets to manufacture reinforcement steel of various sizes with installed capacity of 0.7 million tonnes per year. The electricity in the industry is used to operate the heavy motors for rolling mill, shearing billet, operating pumps, compressors, and blowers while fossil fuels are used in furnace. The industries produced 0.4 million tonnes of product utilizing 59.9 million kWh electricity and 590.3 million MJ energy from different fuel source. The inventory of the energy use is presented in Tables A2, A4 and A6 of the supplementary file.

Coal and firewood are the major sources of energy for brick industries and the electricity is supplied from the national grid for lighting, water pumping and office operation activities. The electricity in the brick industries is mostly supplied through Nepal Electricity Authority's (NEA) supply system. The major source of energy for brick industries is utilized in firing the brick while most other processes are manually done. Hand moulding of bricks is widely practiced in Nepal with natural draft kiln system. The production step includes excavation, clay preparation, brick moulding, sun drying, staking at fireplace, firing, and removing from fireplace, all of which are carried out by human labour except firing. The brick firing is done using coal and firewood. The assessed industries have a total production capacity of 150 million brick per year. The industries produced 144.6 million bricks in the period with energy consumption of 1.8 million kWh electricity and thermal energy of 0.3 million GJ.

2.3 Impact assessment

The energy use for the production material is determined using Eq. (1). The energy use of the material is determined by summating energy use from each source at each step of the process. The emission is determined with the emission factor from each energy source in each step of the production process which is determined using Eq. (2). The energy mix of the country is determined, accounting transmission and distribution losses. The study includes both the electricity and other sources of fuels used in the respective industries.

$$\text{Life cycle energy} = \sum_{i=1}^{n} F_i \times EC_i \tag{1}$$

$$\text{CO2} - \text{eq.} = \sum_{i=1}^{n} F_i \times EF_i + \sum_{j=1}^{m} P_j \tag{2}$$

where, i is the type of energy used, n is the number of energy source used in lifecycle production, F_i is the amount of i energy used in life cycle production of construction material, EC_i is the specific energy content of i energy source, EF_i is the emission factor in the context of Nepal industry transportation/ transmission and distribution losses, m is the type of processes related to material production, P_j is emission from j production process, and m is the type of j process.

Carbon dioxide is generated in the production of cement, brick, and reinforcement steel directly and indirectly. The determination of emission in the production of cement is based on the CO_2-eq. emission factors (see Eq. 2). The emission factor includes process-based CO_2-eq. emission and combustion-based CO_2-eq. emission [46]. Carbon dioxide is indirectly generated in the process from the energy used for the extraction and transportation of raw material and electricity consumed by motors and equipment. The Eco-invent Version 3.9.1 database is used to access the emission from the transportation of the materials from source to the production site. The emission of GHGs is measured as global warming potential (GWP_{100}) of IPCC 2021 with impact measured in kg CO_2-eq. The emission factor calculated is discussed in Sect. 3 and the details are presented in Tables A1 to A7 of the supplementary file.

2.4 Interpretation

Interpretation is the final stage of the LCA study. This phase classifies, evaluates and discusses the outcome of life cycle inventory and impact assessment [47]. It also includes the sensitivity analysis of the study's result and examines scenario analysis. In this phase, conclusions are drawn, and the recommendations may be made based on the result of the developed inventory and impact assessment [48]. The interpretation of the study is discussed in the Sect. 3.

2.5 Scenario analysis for changing energy source

The GoN has planned to generate 15,000 MW hydroelectricity by 2030 from the current state of generation of 1,400 MW [18]. With this the hydroelectricity generation in the country will increase plummeting the imported electricity from India. With the self-sufficiency in electricity the beneficial electrification looks extremely promising. To align with the GoN self-sufficiency target, different scenario was established to study the gradual transition from fossil-fuel based energy use to cleaner energy use in the. Table 2 describes the different scenarios for the energy utilization in material production where gradual transition from fossil-fuel based energy use to beneficial electrification is considered. The base case scenario is the current energy mix of Nepal. Nepal's electricity mix consists of 82.51percent electricity generated from hydropower of Nepal and 17.49 percent of imported electricity from India with overall transmission and distribution loss of 17.18percent [49]. The emission factor for Nepal's energy mix is calculated to be 0.45 kgCO_2-eq./kWh, detail calculation in Table A2 of supplementary file.

3 Results and discussion

The production of cement, brick and reinforcement steel use significant quantity of the fossil fuel mainly coal. Use of electricity in cement and brick production is significantly low as they only use 6.90 percent and 1.86 percent of the energy from electricity respectively. Steel industries use 20.40 percent of the energy from electricity. The industries use grid-connected electricity while the electricity back up is provided by diesel generators for power cut-off. The electricity supplied through the national grid in Nepal comprises 82.51 percent of electricity from the hydropower plants of Nepal while 17.49 percent is imported from India [49].

The emissions from the Nepal's electricity mix are determine based on the emissions from domestically produced electricity and emissions from electricity imported from India. The transmission and distribution loss for the electricity in Nepal is 17.18percent [49]. The calculated emission of the electricity mix of Nepal is 454.05 kg CO_2-eq. Detailed emission data from the electricity mix is presented in the Table A1 of the supplementary file and the transportation emissions factors are presented in Table A2 of the supplementary file. Since coal is the major source of energy in the building material production in Nepal and diesel is predominantly used for transportation, the emission factor for the coal and diesel are calculated at 3.60 kg CO_2-eq per kg of coal and 3.75 kg CO_2-eq per litre of diesel. These factors include emissions from the coal extraction process, transportation from coal mine in India to Nepal, and emissions from the combustion of the coal. The detail calculation of the emission factor of diesel and coal is presented in Tables A3 and A4 of the supplementary file, respectively. Similarly, the emission factor for fuel oil (FO), LPG and firewood are calculated as 4.07 kg CO_2-eq per litre, 2.35 kg CO_2-eq per kg and 1.91 kg CO_2-eq per kg respectively, with details presented in the Tables A5, A6 and A7, respectively.

3.1 Embodied energy

The result of the study on energy use in the production of three major construction material of a modern frame structure buildings is presented in Table 3. The average consumption of electricity to produce one metric ton of cement is 148.6 kWh of which 78.04 percent is supplied from the national grid line while the remaining 21.96 percent is generated from

Table 2 Scenarios of energy transition in material production

Scenario	Description
Base case	Present scenario, with current energy mix of Nepal
I	Scenario I examine the CO_2-eq. emission from the material production after replacing energy generated from fossil-fuel based imported electricity from India by self-generated hydroelectricity of Nepal
II	Scenario II examines the CO_2-eq. emission from the material production while replacing 25% of the energy generated by coal with self-generated hydroelectricity of Nepal
III	Scenario III examines the CO_2-eq. emission after replacing 50% of energy generated by coal in material production with hydroelectricity generated in Nepal as source of energy
IV	Scenario IV examines the CO_2-eq. emission after complete replacement of the energy used in material production by self-generated hydroelectricity in Nepal. This scenario is the case of beneficial electrification

Table 3 Summary of average embodied energy presented in MJ

Source of energy		Cement (per tonne)	Brick (per 1000 pcs)	Reinforcement steel from Billet (per tonne)
Territorial Energy use	Electricity—Grid	417.35	76.57	434.52
	Diesel	196.46	96.90	493.52
	Furnace oil (FO)	26.00	-	663.1
	Coal	5411.26	3863.0	462.72
	Firewood	–	88.09	4.66
	Rice husk	–	–	69.21
	LPG	–	–	2.48
Energy use on ore to billet production [50]		-	–	23,472.24
Transportation from billet industry to reinforcement steel industry		-	–	430.69
Total		6051.07	4124.56	26,033.14

diesel generators. Similarly, the energy of 5411.26 MJ is sourced from Coal and 26.0 MJ from fuel or furnace oil (FO) in the production of one metric ton of cement. The average total embodied energy from all sources of energy to produce one ton of cement is equivalent to 6051.07 MJ. The summary of the energy use in cement production is presented in Table 3 and the detail is presented in Table A8 of the supplementary file.

The brick production industries use human resources for most of the unit process and use coal for the firing of the brick. The moulded bricks are sun-dried to evaporate the moisture contained in the brick before the firing. The total embodied energy in the production of 1000 pieces of brick is 4124.56 MJ of which firing of the brick consumes 3951 MJ (95.79%) of energy. Of the energy consumed 3863.0 MJ (93.66) is sourced from coal, 88.09 MJ is supplied from firewood, 96.90 MJ from diesel, and the remaining from Grid-connected electricity. The electricity consumption to produce 1000 pieces of brick is 21.27 kWh. The brick industries also use agricultural residue to replace coal and firewood when available locally. The study on energy use in the production of brick in different parts of the world ranges from 0.36 to 10.02 MJ/kg [45] while the energy use in the production of brick in Nepal remains on the lower side with just 2.06 MJ/kg. This is because most of the activities in the production of brick are done manually by humans and transportation inside the factory is done by humans or donkeys, as 1349 brick industry employed 246,000 people and 3300 animals in 2018 for the production of 5.1 billion bricks [19]. The summary of the energy use in brick production is presented in Table 3 and the detail is presented in Table A10 of the supplementary file.

The production of one functional unit (one tonne) of the reinforcement steel from billet consumes 2130.2 MJ of energy from different energy sources like coal, FO, diesel, and electricity. The process utilizes 149.2 kWh of electricity of which 120.7 kWh is supplied from the national grid and 28.5 kWh is from diesel generators. Similarly, 462.72 MJ of energy is supplied from coal and 663.1 MJ from fuel oil. Similarly, 76.35 MJ of energy is supplied from firewood, rice husk, and LPG. The energy used in production of billet from iron ore is 23, 472.24 MJ [50] with transportation accounting 430.69 MJ of energy. The total energy use in production of reinforcement steel is 26,033.14 MJ. The summary of the energy use in brick production is presented in Table 3 and the detail is presented in Table A12 of the supplementary file.

3.2 Carbon emissions

The result of the study on CO_2-eq emission associated with the energy use in the production of three major construction material of a modern frame structure buildings is presented in Table 4. Cement industries emit CO_2-eq. from the combustion of the fuel for different production processes, that involves transportation, energy use and calcination [1]. The total CO_2-eq. equivalent emission to produce one tonne of cement is 739.5 kg CO_2-eq. The emission from coal alone accounts for 664.9 kg CO_2 equivalent while electricity and diesel account for 52.6 kgCO_2-eq. and 19.4 kg CO_2-eq., respectively. According to the IPCC 2021 climate change global warming potential (GWP_{100}) from the ecoinvent 3.9.1 Dataset Documentation, the GWP_{100} for tonne of Portland cement production is 735.5 kg CO_2-eq. for Switzerland, 887.9 kg CO_2-eq. for India and 910.2 kg CO_2-eq. for rest of the world [51]. The lowest emissions from the cement production is observed in China, at 460 kg CO_2-eq which might be attributed to a lower clinker-to-cement ratio and energy-efficient kilns [52]. The emission from cement production in Nepal is relatively higher in comparison to Switzerland due to the use of coal,

Table 4 Summary of average energy use associated carbon emission in $kgCO_2$-eq

Source of Energy		Cement (per tonne)	Brick (per 1000 pcs)	Reinforcement steel from Billet (per tonne)
Territorial emission	Electricity – Grid	52.64	9.66	54.80
	Diesel	19.39	9.56	48.70
	Furnace oil (FO)	2.58	–	63.26
	Coal	664.88	474.65	56.85
	Firewood	–	9.02	0.48
	Rice husk	–	–	8.79
	LPG	–	–	0.12
Emission from ore to billet production [50]		–	–	2290.00
Transportation from billet industry to reinforcement steel industry		–	–	42.50
Total emission ($kgCO_2$-eq.)		739.49	502.89	2565.51

while it's lower in comparison to India and an average of the rest of the world. This is due to cleaner sources of electricity from hydropower, while in India and other developing countries the major source of energy for electricity is fossil fuels.

The primary method for brick firing is the combustion of coal and the burning of firewood. The total carbon dioxide equivalence generated from the production of 1000 pieces of burnt bricks is 502.9 kg (i.e., 0.503 kg CO_2-eq. per brick). Of this total emission, 474.7 kg of CO_2-eq. is generated from the combustion of coal. Coal accounts for 94.4 percent of the total emissions while electricity accounts for 1.9 percent, rest is from firewood which is mostly considered as biogenic CO_2. Studies on emissions from brick production in different parts of the world range from 0.01 to 0.33 kg CO_2-eq./kg [45]. According to IPCC 2021 GWP_{100} based on ecoinvent 3.9.1 Dataset documentation, the global emission factor is 0.33 kg CO_2-eq [51]. The emission from Nepal remains in the upper range with an emission of 0.25 kg CO_2eq./kg. Although the energy use remains on the lower side in comparison to other part of world, the emission rate is higher rate due to the use of high carbon emitting fuel like coal. The high carbon emitting coal is further imported from India imparting the emission during emission.

The reinforcement steel production from billets accounts for the emission of 233.01 kg CO_2-eq. per tonne. The billet production from the iron ore in Tata Steel Jamshedpur industry is 2290.0 kg CO_2-eq. per tonne [50]. The emission from transportation of one tonne of billets from the Jamshedpur plant to manufacturing industry in Nepal accounts for 42.5 kg CO_2-eq. The cumulative emission of manufacturing of reinforcement steel from ore is calculated to be 2565.5 kg CO_2-eq. per tonne of reinforcement steel. According to the IPCC 2021 climate change GWP_{100} from the ecoinvent 3.9.1 dataset documentation, the emissions for steel production are 2180 kg CO_2-eq. per tonne of steel for the rest of the world [51]. While the results from the LCA study conducted in Italy show emissions of 1600 kg CO_2-eq per tonne of steel [33]. The higher emission of Nepal in comparison to the rest of the world is due to higher use of coal as energy source. The summary of the emission from building material production is presented in Table 4 and the detail on energy use and emission from different phase is presented in Table A15 of supplementary file.

The use of traditional technology and coal as a major source of energy in the industries is the major reason for higher emissions in the country. Similarly, the measure to enhance the energy efficiency of the industries shall have the potential to minimize energy use and emission. The measure to promote the recycling of the metal shall minimize the use of imported billet in the production of reinforcement steel shall reduce both direct and indirect emission and energy use.

As the country has made the commitment to achieve net zero emission by 2045 and made a specific commitment in the second NDC to adopt low-emission technologies in brick and cement industries to reduce coal consumption, shall influence the reduction of embodied energy of cement and brick in soon near future. Similarly, increasing the use of efficient machines and automation in the production process shall minimize the energy use in the production of all three cement, brick, and reinforcement steel, further reducing the emission from the material production. Similarly, the study shows that natural aggregate is the another most emitting materials in contemporary reinforced concrete building [14], the study on energy use and emission from production of aggregate shows that production of one tonne of crushed natural aggregate utilize 55.83 MJ of energy use and generate emission of 4.45 CO_2-eq/ tonne [53].

Reduction in the use of higher emitting materials in construction activities with their replacement with natural materials like wood and stone shall minimize the emission from the built environment. The energy use and emission from

ecoinvent 3.9.1 Dataset documentation for the processing and production of the stone for construction work is just 90.1 MJ/m^3 and emits 7.9 kg CO_2-eq /m^3 while that for wood is 545.8 MJ/m^3 and 45.7 kg CO_2-eq /m^3. This suggests that increasing the use of wood and stone in the construction work and replacing steel bars, cement and brick shall reduce the energy demand and emission from the built environment sector supporting the long-term goal of net zero emissions.

3.3 Sensitivity analysis for emission

The study shall have some level of uncertainty because of the data availability and the emission inventory of the country. The major determining factor of CO_2-eq. emission in the production of cement, brick and reinforcement steel is the source of energy use. As coal is the major source of energy and one of the main contributing factors in the production of all three construction materials, sensitivity analysis was conducted with differing amounts of coal in the production of all three cement, brick, and reinforcement steel. Coal accounts for 89.9 percent of energy in the cement production process, reducing the use of coal by 10 percent in the production of cement will decrease the overall emission by 8.99 percent. Similarly in brick production coal accounts for 94.4 percent of the total emission, so minimizing the use of coal by 10 percent in brick production will reduce the emission by 10.4 percent of the total emission. The emission in the production of reinforcement steel from steel billets in Nepal is influenced by diesel, electricity, coal, and fuel oil, so trimming down these sources of energy by 10 percent will cut the territorial emission by 2.1 percent, 2.4 percent, 2.44 percent, and 2.7 percent of the total emission respectively. The change in emission due to change in energy quantity by 10 percent sensitivity for reinforcement steel is presented in Fig. 2.

As coal is the major source of energy and emissions in the building materials production, changes in the emission factor of the coal have the potential to minimize the embodied emissions of the building material. The sensitivity analysis was conducted to evaluate the impact of changes in the coal emission factor. A 10 percent change in the emission factor of coal will result in an 8.99 percent change in the total emissions from cement production, a 9.44 percent change in total emissions from the brick production, and a 2.44 percent change in territorial emissions from reinforcement steel. Technological improvement and energy efficiency shall reduce the emission factor contributing to the overall reduction in the emission from the material production [13, 54].

3.4 Scenario analysis for changing energy source

The sensitivity analysis of the study shows that there is a significant reduction in carbon emissions when the energy mix in the production process is gradually changed to beneficial electrification. Four scenarios were developed as presented in Table 2 to assess the change in emission from the production cement, brick, and reinforcement steel. The four scenarios were established considering the changing electricity mix of the country and GoN target of minimizing coal use in industries. The Figs. 3, 4, 5 and 6 depicts emission and energy scenario for different material production in different scenarios. Achieving self-sufficiency in electricity generation in Nepal and achieving scenario I of complete use of self-generated hydroelectricity in Nepal, will lead to a 6.7 percent reduction in emission of cement from present scenario. Likewise, transitioning to scenario II of while replacing 25 percent of the energy generated by coal with self-generated

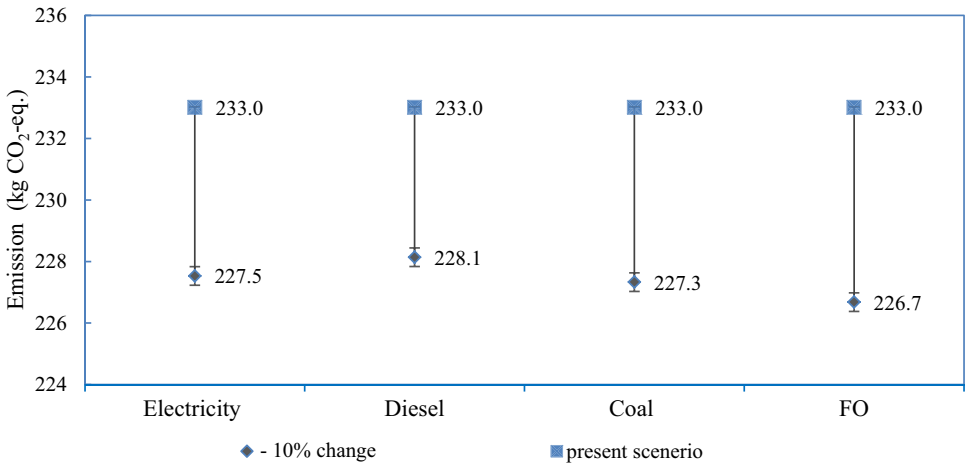


Fig. 2 Emission cut-off from reinforcement steel production with 10% reduction of energy quantity

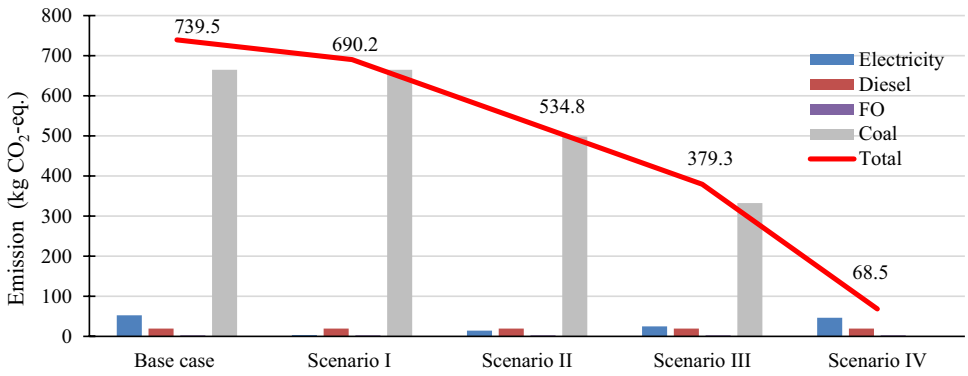


Fig. 3 Emissions from cement production (per tonne) with changing energy source from coal to hydroelectricity

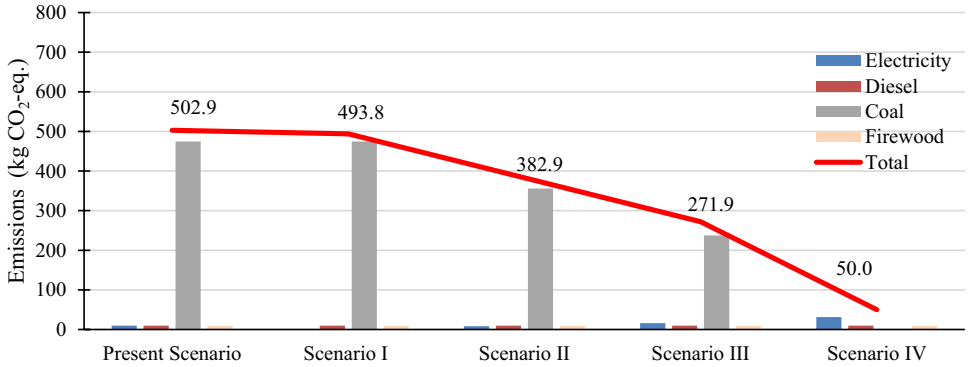


Fig. 4 Emissions from brick production (per 1000 pieces) with changing energy source from coal to hydroelectricity

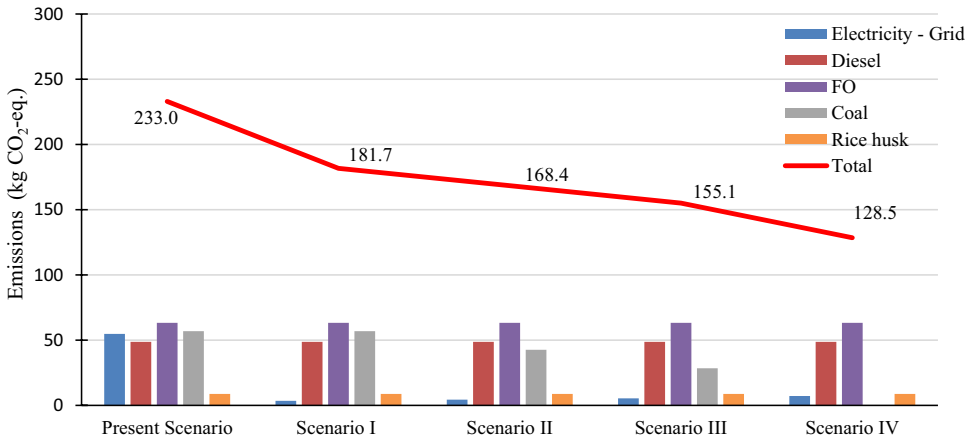


Fig. 5 Emissions from reinforcement steel production (per tonne) with changing energy source from coal to hydroelectricity

hydroelectricity of Nepal, will result in 27.7 percent emission reduction. Furthermore, adopting scenario III of replacing 50 percent of energy generated by coal in material production with hydroelectricity generated in Nepal as source of energy, will lead to a substantial 48.7 percent reduction in emissions. While on achieving scenario IV of complete replacement of the energy used in material production by self-generated hydroelectricity in Nepal, the emission will be reduced drastically. The reduction of emission from energy transition from coal to electricity is due to lower emission associated with hydropower generated electricity. The scenario of emission from shifting energy source from coal to cleaner hydropower generated electricity in cement production is presented in Fig. 3 and the data is presented in Table A9 of supplementary file.

Transiting from base case scenario to Scenario I will reduce the emission of brick production by only 1.80 percent, this is lower than cement and reinforcement steel production due to lower electricity use as a source of energy in brick production. Similarly, achieving scenario I of shifting the source of energy from coal to electricity generated in Nepal by 25 percent shall reduce the emission by 23.9 percent. While achieving scenario II, replacement of 50 percent of the

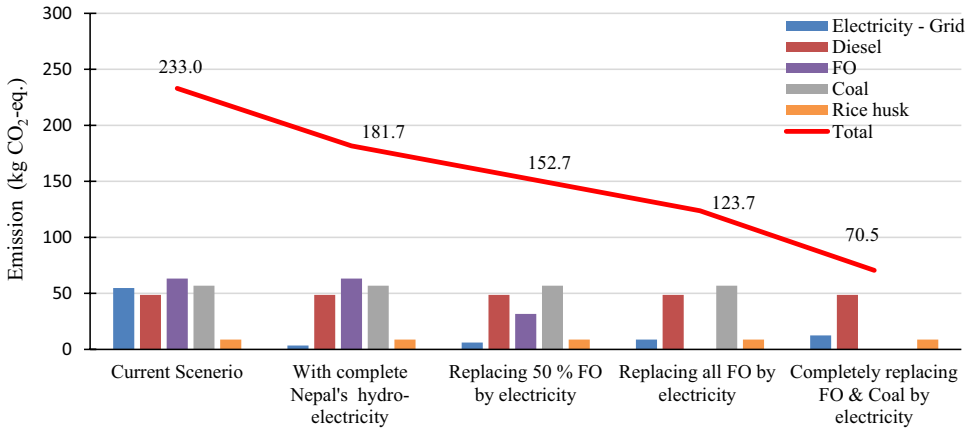


Fig. 6 Emissions from reinforcement steel production (per tonne) with changing energy source from FO to hydroelectricity

coal by electricity, shall reduce the emission by 45.9 percent. The scenario of emission from shifting energy source to cleaner hydropower electricity in brick production is presented in Fig. 4 and the data is presented in Table A11 of supplementary file.

With self-sufficiency in electricity generation, achieving scenario I shall reduce the territorial emission of reinforcement steel production by 22.0 percent which is higher in comparison to the reduction of emission in production of cement and brick. Similarly, achieving scenario II will curb the emission by 27.7 percent and achieving scenario III will curtail the emission by 33.4 percent. The scenario of emission from shifting energy source to cleaner hydropower generated electricity from coal in reinforcement steel production is presented in Fig. 5 and shifting from FO to hydropower generated electricity is presented in Fig. 6 and the data for both the scenario is presented in Tables A13 and A14 of supplementary file.

With the increasing technological dissemination, human involvement is plummeting in cement, brick, and reinforcement steel production activities and the energy use in the production process shall increase in the future. However, the increasing use of electricity replacing fossil fuel due to the sufficiency of electricity in the country and its reliability shall reduce the emissions.

3.5 Limitation

The study presents the embodied energy and the associated carbon emissions for the three major building materials. The construction material production process in Nepal uses a considerable number of human resources. However, the study was not able to account for the human resources and their transportation involvement in material production. The future study could include direct emissions associated with the production process which was beyond the scope of this paper.

4 Conclusion

The study shows that Cement, brick, and reinforcement steel have high embodied energy in their production and contribute substantial amounts of CO_2-eq emissions. The major sources of energy used in the production of building materials in Nepal is coal, while the use of electricity in the material production is significantly low. The electricity supplied through the national grid comes from hydropower or is imported from India. The use of electricity in reinforcement steel is relatively higher compared to that in cement and brick production. Transitioning from coal to hydropower generated electricity is essential to minimize the energy related emissions from the building material production in Nepal. This shift would also ensure the effective utilization of renewable energy generated in the country, aligning with the government of Nepal's target outlined in its Second Nationally Determined Contribution. Similarly, the shift in building construction practices in Nepal from traditional structures to reinforced concrete frame structures is increasing the use of cement, fired bricks, and reinforcement steel, with limited study on the associated environmental impacts and emissions. Therefore, Nepal's building sector needs to adopt strategies to minimize environmental impact and promote the circularity of these materials at the end-of-life stage through reuse and recycling. Considering the higher embodied energy and emission of the contemporary building materials, the production of these materials needs to be carefully addressed in strategies for achieving the national and global goal of GHG reduction. Effective formulation and implementation of

policies to regulate energy use and emissions, along with strict monitoring are necessary to mitigate CO_2-eq emissions from building material production. Also, reducing the use of these high emitting construction materials in buildings by replacing them with locally available natural materials like stone, slate, wood and bamboo will minimize the emissions from the built environment as well as use of these natural material in the building makes the building more adopted to the local climate conditions.

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Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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